

Transient Reactance

X' (transient reactance) is the ratio of no-load emf (E_n) to the transient symmetrical r.m.s current.

Subtransient Reactance

X'' (subtransient reactance) is the ratio of no-load emf (E_n) to the subtransient symmetrical r.m.s current.

Application of Transient/Subtransient Reactance

The transient and subtransient reactances help in calculating the interrupting and maximum momentary short-circuit currents.

Per Unit:

Per Unit (pu) value of any quantity is defined as the ratio of actual value of the quantity to the base value of the same quantity.

$$\text{Normally the nominal voltage of lines and equipment is known along with the complex power rating. Hence, in practice, the base values are chosen for complex power, Mega-Volts-Amperes (MVA) and line voltage, kilo-Volts (KV). The chosen base (MVA)_b is the same for all the parts of the network. However, the base voltage (KV)_b is chosen with reference to a particular section of the network and the other base voltages (with reference to the other sections of the network) due to the presence of transformer are then related to the chosen one by the turns-ratio of the connecting transformer.$$

Let I_b be the base current in kilo-Amperes (KA)_b and V_b the base voltage in (KV)_b, then base complex power $S_b = (MVA)_b$

$$\text{Then } S_b = V_b \cdot I_b \quad (22)$$

$$\text{Also } (MVA)_b = (KV)_b \cdot I_b \quad (25)$$

$$\text{The base current } (KA)_b = I_b = \frac{(MVA)_b}{(KV)_b} = \frac{S_b}{V_b} \quad (29)$$

(1)

The Base Impedance, $Z_b = \frac{V_b}{I_b} = \frac{(KV)_b}{I_b}$ (30)

From eqn (29) $I_b = \frac{(MVA)_b}{(KV)_b}$ and substituting for I_b into eqn (30) gives;

$$Z_b = \frac{(KV)_b}{(MVA)_b / (KV)_b} = \frac{(KV)_b}{1} \times \frac{(KV)_b}{(MVA)_b} = \frac{(KV)_b^2}{(MVA)_b}$$

Hence,

$$Z_b = \frac{(KV)_b^2}{(MVA)_b} \quad (31)$$

Per unit (pu) impedance is given;

$$Z_{pu} = \frac{\text{Actual } Z_{(ohms)}}{Z_b} \quad (32)$$

Substituting for Z_b in eqn (32) gives;

$$Z_{pu} = \frac{Z_{(ohms)}}{(KV)_b^2 / (MVA)_b} = \frac{Z_{(ohms)}}{1} \times \frac{(MVA)_b}{(KV)_b^2}$$

Hence, $Z_{pu} = \frac{Z_{(ohms)} \times (MVA)_b}{(KV)_b^2} \quad (33)$

In conclusion, in 3-phase networks, $(KV)_b$ is the line-to-line voltage and $(MVA)_b$ is the 3-phase MVA. [1-phase MVA = one third of 3-phase MVA]

Change of Base:

To change from old Z_{pu} to new Z_{pu} then,
 New $Z_{pu} = \text{Old } Z_{pu} \left[\frac{\text{New } (MVA)_b}{\text{Old } (MVA)_b} \right] \left[\frac{\text{Old } (KV)_b}{\text{New } (KV)_b} \right]^2 \quad (34)$

Example 1:

A generator rated at 30 MVA, 11KV has a reactance of 20%. Calculate its p.u reactance for a base of 50MVA and 10KV.

$$\text{Old (KV)}_b = 11, \text{ New (KV)}_b = 10$$

$$\text{Old (MVA)}_b = 30, \text{ New (MVA)}_b = 50$$

$$\text{old } X_{p.u} = 20\% = 0.2, \text{ New } X_{p.u} = ?$$

$$\begin{aligned} \text{New } X_{p.u} &= \text{Old } X_{p.u} \left[\frac{\text{Old (KV)}_b}{\text{New (KV)}_b} \right]^2 \left[\frac{\text{New (MVA)}_b}{\text{Old (MVA)}_b} \right] \\ &= 0.2 \left[\frac{11}{10} \right]^2 \left[\frac{50}{30} \right] = 0.403 \text{ p.u} \end{aligned}$$

Example 2:

The base KV and base MVA of a 3-phase transmission line is 33KV and 10 MVA respectively. Calculate the base current and base impedance.

$$(KV)_b = 33, (MVA)_b = 10$$

$$\text{Base current } I_b = \frac{(MVA)_b}{(KV)_b}$$

$$= \frac{10 \times 10^6}{33 \times 10^3}$$

$$= \frac{10 \times 10^3}{33}$$

$$= \frac{10,000}{33}$$

$$= \underline{\underline{303.03 \text{ A}}}$$

$$\text{Base impedance } Z_b =$$

$$\frac{(KV)_b^2}{(MVA)_b}$$

$$= \frac{(33 \times 10^3)^2}{10 \times 10^6}$$

$$= \underline{\underline{108.9}}$$

(3)

Example 8:

A 100 MVA, 13.8 kV, Y-connected, 3-phase 50 Hz synchronous generator is operating at the rated voltage and no-load, when a 3-phase fault occurs at its terminals. Its reactances per unit to the machine's own base are, $X_{se} = 1.00$, $X' = 0.25$ and $X'' = 0.12$. Calculate the subtransient, transient, and steady-state currents for the fault.

$S_b = 100 \text{ MVA}$, $V_b = 13.8 \text{ kV}$, Hence no-load voltage p.u., $E_k(\text{p.u.}) = \frac{13.8 \text{ kV}}{13.8 \text{ kV}}$

$$I_{b(\text{line})} = \frac{S_b}{\sqrt{3} V_b(\text{line})} = \frac{100 \times 10^6}{\sqrt{3} \times 13.8 \times 10^3} = 1.0 \text{ p.u.}$$

$$= \underline{4184 \text{ A}}$$

$$\text{Subtransient current } I'' = \frac{E_k}{X''} = \frac{1.0}{0.12} = 8.333 \text{ p.u.}$$

$$\text{Actual } I'' = 8.333 \times 4184 = \underline{34900 \text{ A}}$$

$$\text{Transient current } I' = \frac{E_k}{X'} = \frac{1.0}{0.25} = 4 \text{ p.u.}$$

$$\text{Actual } I' = 4 \times 4184 = \underline{16700 \text{ A}}$$

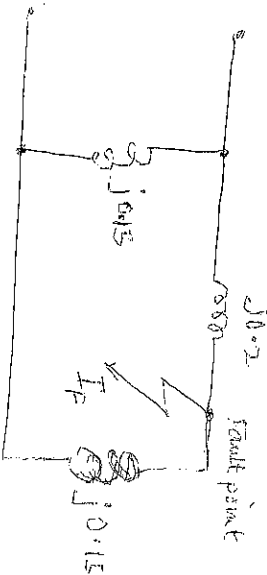
$$\text{The steady-state current, } I_{ss} = \frac{E_k}{X_{es}} = \frac{1.0}{1.0} = 1 \text{ p.u.}$$

$$\text{Actual } I_{ss} = 1 \times 4184 = \underline{4184 \text{ A}}$$

(4)

Example 4:

Find the fault current, I_f in the figure below if the pre-fault voltage at the fault point is 0.97 p.u. ?



Pre-fault voltage $V_{pf} = 0.97$

Hence,

$$I_f = \frac{V_{pf}}{Z_{th}} = \frac{0.97}{j0.2 + j0.15} = j0.105 \text{ p.u.}$$

$$\text{Fault Current, } I_f = \frac{V_{pf}}{Z_{th}} = \frac{0.97}{j0.105}$$

$$= j9.238 \text{ p.u.}$$

Faults:

Fault in a circuit is any failure which interferes with the normal flow of current. The faults are associated with abnormal change in current, voltage and frequency of the power system. Faults occur in power network due to:

- (1) insulation failure of equipment,
- (2) flashover of lines initiated by lightning stroke,
- (3) due to permanent damage to conductors and towers or due to accidental faulty operation,
- (4) falling of a tree along the line,
- (5) wind and ice loading on the transmission lines,
- (6) vehicles colliding with supporting structures,
- (7) overloading of underground cables,
- (8) Birds shorting lines.

(*)

- (9) Lightning
- (10) Dew, fog, frost.
- (11) snow
- (12) gales
- (13) salt spray.

List of various types of short and series faults

Short faults are;

- (1) Line to ground (L-G) fault.
- (2) Line to line (L-L) fault.
- (3) Double to ground (L-L-G) fault.
- (4) Three phase fault (L-L-L) fault.

Series faults are;

- (1) One open conductor fault,
- (2) Two open conductor fault,
- (3) Three open conductor fault.

Frequency of Fault Occurrence

- (1) Three phase fault 5%
- (2) Line to line to ground fault 10%
- (3) Line to line fault 15%
- (4) Single line to ground fault 70%

Fault (Short Circuit) Analysis

An essential part of the design of a power supply network is the calculation of the currents which flow in the components of the system when faults of various types occur. The magnitudes of fault currents give the engineer the current settings so as to enable him to determine the most adequate protection to be used and the ratings of fuses, circuit breakers, switchgears, busbars in addition to setting up protective relays. Hence, the purpose or objective of fault (or short circuit) analysis is as follows;

- (1) To determine maximum and minimum three-phase short circuit currents.
- (2) For selecting fuses, circuit breakers, switchgears, busbars and protective relays of adequate making and breaking capacities.
- (3) For selecting busbar supports and cables capable of withstanding thermal and mechanical stresses because of short-circuit.

- (4) To do current based discrimination between circuit breakers.

Fault Level

The MVA ^{fault} level and fault current, I_f are frequently used ^{as} the ratings of the protective devices. The short circuit capacity or the short circuit or fault level at a bus is defined as the product of the magnitude of the rated bus voltage and the fault current. That is, the fault MVA level = $V_L I_f \sqrt{3} \times 10^6$, where

V_L is the nominal line voltage of the faulted part. The fault MVA is often referred to as the "fault level". Fault level at any given point of the electrical power supply network is the maximum current/power that would flow in case of a short circuit fault at the point.

Short circuit

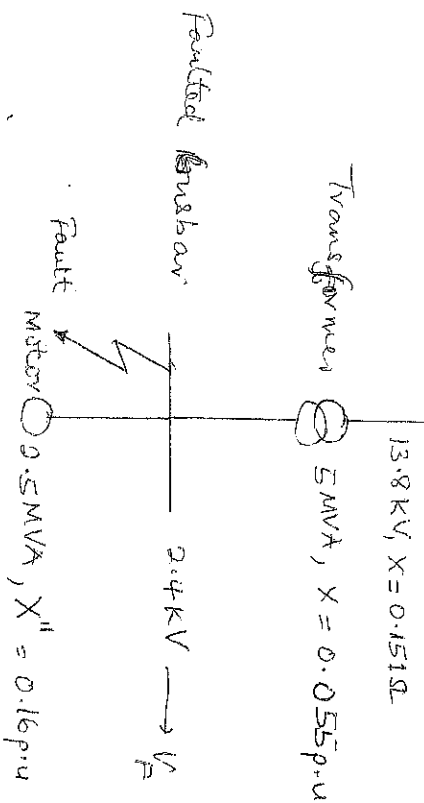
A short circuit is said to have occurred whenever, such a fault causes a large current to flow in one or more phases. Fault currents cause equipment damage due to both thermal and mechanical stresses.

Methods of Estimating Fault Levels

There are various analytical methods of estimating fault levels in power systems. The commonly used methods are Ohmic, per-unit and MVA methods. Ohmic and per-unit methods are cumbersome and mistakes can be easily made because of conversions and fractions involved. The MVA method is simpler and considered fine.

Example

Determine the fault level at 2.4 kV busbar of the network below.
System (Utility) 500 MVA



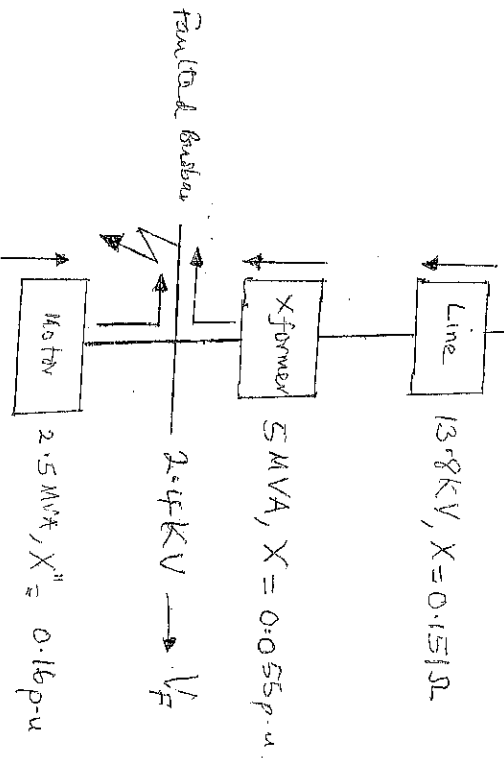
Single Line Diagram of Power Network

Solution

In solving the problem the following steps are followed by drawing the single diagram to block diagram as shown below.

(7)

$$\text{System } 500 \text{ MVA} \quad (MVA)_S = 500$$



Power as actual $P = \frac{V^2}{R} = \frac{V^2}{X}$

$$\text{Line } (MVA)_L = \frac{(kV)_L^2}{X_L} = \frac{(13.8 \times 10^3)^2}{0.151} = 1261 \text{ MVA}$$

$$\text{Transformer } (MVA)_T = \frac{(MVA)_T}{X_{T(p.u.)}} = \frac{5}{0.055} = 91 \text{ MVA}$$

$$\text{Motor } (MVA)_M = \frac{(MVA)_M}{X_{M(p.u.)}} = \frac{2.5}{0.16} = 15.6 \text{ MVA}$$

Fault $(MVA)_F$ level at 2.4 kV busbar without considering the motor is,

$$\begin{aligned} (MVA)_F &= \frac{1}{(MVA)_S} + \frac{1}{(MVA)_L} + \frac{1}{(MVA)_T} \\ &= \frac{1}{500} + \frac{1}{1261} + \frac{1}{91} \\ &= 0.002 + 7.93 \times 10^{-4} + 0.010989 \\ &= 0.0137826 \end{aligned}$$

$$\therefore (0.0137826) \times (MVA)_F = 1$$

$$\therefore (MVA)_F = \frac{1}{0.0137826} = \underline{\underline{72.56 \text{ MVA}}}$$

Hence, fault current $I_F =$

$$\begin{aligned} &= \frac{(MVA)_F}{\sqrt{3} \times V_F} \\ &= \frac{72.56}{\sqrt{3} \times 2.4 \times 10^3} \\ &= \underline{\underline{17.48 \text{ kA}}} \end{aligned}$$

If the motor is considered, the (MVA) fault level is calculated as follows;

$$\begin{aligned} (MVA)_{\text{with motor}} &= (MVA)_{\text{without motor}} + (MVA)_M \\ &= 72.56 + 15.6 \\ &= 88.16 \text{ MVA} \end{aligned}$$

Hence the fault current with motor $I_{F(\text{with motor})}$ is

$$\begin{aligned} I_{F(\text{with motor})} &= \frac{(MVA)_{(\text{with motor})}}{\sqrt{3} \times V_F} \\ &= \frac{88.16 \times 10^6}{\sqrt{3} \times 2.4 \times 10^3} \\ &= \underline{\underline{22.23 \text{ kA}}} \end{aligned}$$

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